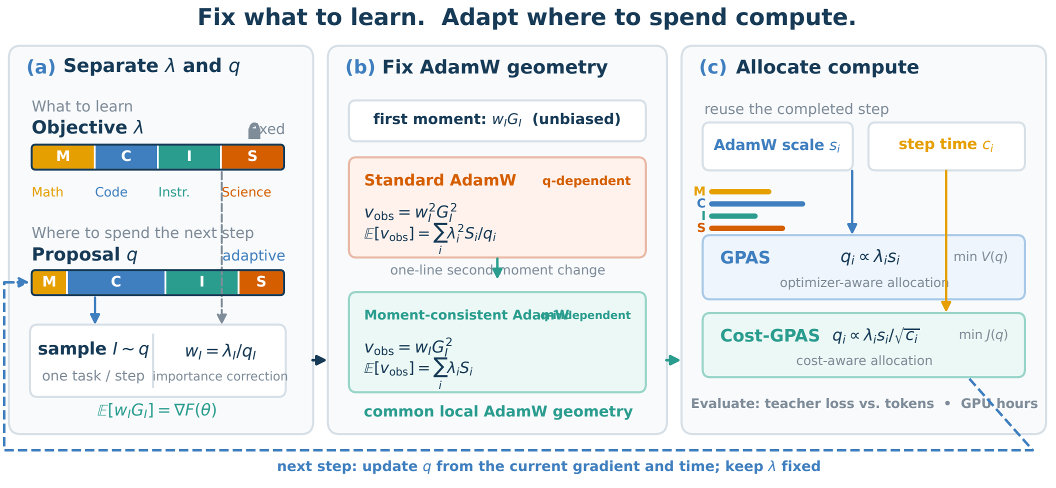

Fix what to learn. Adapt where to spend compute.
(a) Separate λ and q
What to learn
Objective λ
fixed
M
C
I
S
Math
Code
Instr.
Science
Where to spend the next step
Proposal q
adaptive
M
C
I
S
sample $l \sim q$
one task / step
$w_l = \lambda_l / q_l$
importance correction
$\mathbb{E}[w_l G_l] = \nabla F(\theta)$
(b) Fix AdamW geometry
first moment: $w_l G_l$ (unbiased)
Standard AdamW
q-dependent
$v_{\text{obs}} = w_l^2 G_l^2$
$\mathbb{E}[v_{\text{obs}}] = \sum_i \lambda_i^2 S_i / q_i$
one-line second moment change
Moment-consistent AdamW
q-independent
$v_{\text{obs}} = w_l G_l^2$
$\mathbb{E}[v_{\text{obs}}] = \sum_i \lambda_i S_i$
common local AdamW geometry
(c) Allocate compute
reuse the completed step
AdamW scale s_i
step time c_i
M
C
I
S
GPAS
$q_i \propto \lambda_i s_i$
min $V(q)$
optimizer-aware allocation
Cost-GPAS
$q_i \propto \lambda_i s_i / \sqrt{c_i}$
min $J(q)$
cost-aware allocation
Evaluate: teacher loss vs. tokens • GPU hours
next step: update q from the current gradient and time; keep λ fixed